Newt Mac VR Showcase

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A native Godot 4 OpenXR demo for Mac: one small island, three stations, built entirely in code from primitives — no imported assets. It exists to show what Mac-native VR through OpenXR (Godot's Vulkan renderer on a local runtime such as the Monado Quest Link driver) can do today, and to exercise the Newt session-manager handoff end to end.

For the minimal starter project, see `../godot`. This is the bigger sibling.

1 What's in the scene

- **The Atrium** — emissive crystal spire, counter-rotating metallic rings, a GPU particle fountain, procedural sky, and a day/dusk lighting toggle with animated transitions.
- **Physics Yard** — a crate pyramid, bouncy balls, and a seesaw plank. Grab any prop with the grip, carry it, throw it with real hand velocity.
- **Target Range** — five floating targets on animated paths that speed up as you score, ray shooting with haptic pulses and generated sound effects.
- **Control pedestal** — physical 3D buttons: Spawn Ball, Reset Physics, Dusk/Day, and Teleport on/off.
- **Everywhere** — 140-instance MultiMesh scatter in one draw call, dynamic shadows, ACES tonemapping with glow, fog, and a wrist HUD with live score/shots/FPS.

2 Controls

Input	Action
Left thumbstick	Smooth move (or nothing in teleport mode)
Right thumbstick	Snap turn 45°
Right thumbstick ↑ (teleport mode)	Aim teleport arc, release to jump
Trigger	Shoot targets / press 3D buttons
Grip (near a prop)	Grab · release to throw
Menu button	Pause panel: Resume, or Quit to Newt

Quitting follows the Newt cooperative convention: the game exits its process and the session manager restores the Newt dashboard.

Desktop preview (no headset): the camera slowly orbits the island. `Esc` opens the pause panel, `Q` quits from it.

3 Run it

Requirements match the minimal example: Godot 4.4 with OpenXR + Vulkan on macOS, a compatible separately-supplied OpenXR runtime, matching architectures, and an awake headset with no other app owning the session.

Editor Play (recommended): open this folder in Godot, configure Newt once via **Project → Tools → Configure Newt...**, press Play. The bundled Newt Play Bridge starts/reuses the session manager, reserves the headset, and restores the dashboard on Stop.

Session manager CLI:

```
# After building Newt (or use the installed ‘newt‘ command):
../../build/newtd run --project "$PWD"
```

Direct launcher (no Newt):

```
python3 launch.py --runtime /absolute/path/to/runtime.json --godot /path/to/Godot
```

4 Validate without a headset

```
godot --headless --path . --import
godot --headless --path . -s tests/showcase_smoke.gd
```

The smoke test instantiates the real scene in desktop-preview mode and drives every pedestal action, a scoring shot, and the pause flow. It validates scripts and scene assembly only. OpenXR initialization, controller bindings, frame timing, USB transport, and comfort still need a hardware run — that part is honestly unverified until someone puts the headset on.

5 Layout

File	Role
`main.gd`	World building, stations, scoring, pause, day/dusk
`scripts/player_rig.gd`	Hands, rays, locomotion, teleport, grabbing, HUD
`scripts/factory.gd`	Materials, props, 3D buttons, signs
`scripts/sound_bank.gd`	Procedural WAV sound effects (no assets)
`tests/showcase_smoke.gd`	Headless interaction smoke test
`addons/newt_bridge/`	Newt Play Bridge (installed, do not edit here)

Godot owns its OpenXR session here — `Newt_Run()` is never called inside the game. Newt's role is the dashboard/launcher around it and the one-owner handoff, exactly as in the minimal example.